CAD/CAM Quadruped Robot Report

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Abstract

This project presents the design and development of a simplified quadruped robot using CAD modeling, gait analysis, and motion simulation techniques. The robot consists of a rectangular body frame and four identical legs, each having two degrees of freedom (hip and knee joints).

The complete robot assembly was created in SolidWorks, where appropriate mates and joint constraints were applied to achieve realistic leg movement. To study locomotion, a basic gait cycle was generated using mathematical trajectory equations and implemented in Python. Hip and knee joint angle variations were plotted to analyze the walking pattern of the robot. Motion validation was performed through simulation to ensure smooth, collision-free movement and proper leg coordination.

A stability analysis was also conducted to verify that the robot maintains balance during locomotion. The results demonstrate the feasibility of the proposed quadruped design and provide a foundation for future developments involving advanced gait control, sensor integration, and autonomous navigation. This project enhances understanding of robotic locomotion, CAD-based design, kinematic analysis, and simulation techniques used in modern legged robots.

Introduction

Quadruped robots are legged robotic systems that use four legs for locomotion. Unlike wheeled robots, quadruped robots can move efficiently on uneven surfaces, climb obstacles, and maintain stability in challenging environments. Due to these advantages, they are widely used in applications such as industrial inspection, search and rescue operations, military surveillance, exploration of hazardous areas, and autonomous navigation. The development of quadruped robots involves several engineering disciplines, including mechanical design, kinematics, dynamics, control systems, and simulation.

The design of an effective quadruped robot requires proper leg geometry, joint configuration, motion planning, and stability analysis to achieve smooth and efficient locomotion. In this project, a simplified quadruped robot was designed and modeled using SolidWorks CAD software. The robot consists of a rectangular body frame and four identical legs, each having two degrees of freedom (DOF) through hip and knee joints.

A basic gait cycle was generated using mathematical equations and implemented in Python to analyze the variation of joint angles during walking. Motion simulation was performed to validate the movement of the robot and verify the effectiveness of the designed gait pattern. The primary objective of this project is to understand the fundamentals of quadruped robot design, gait generation, motion validation, and stability analysis. The project provides practical exposure to CAD modeling, robotic locomotion concepts, and simulation techniques, which are essential in modern robotics applications.

Objectives

1. To design and develop a simplified quadruped robot using CAD modeling techniques.

2. To create a complete 3D model and assembly of the quadruped robot in SolidWorks.

3. To design four identical legs with two degrees of freedom (hip and knee joints) for locomotion.

4. To define appropriate joint constraints and motion limits for realistic leg movement.

5. To generate and analyze a basic gait cycle for quadruped walking.

6. To develop joint trajectory plots for hip and knee joints using Python.

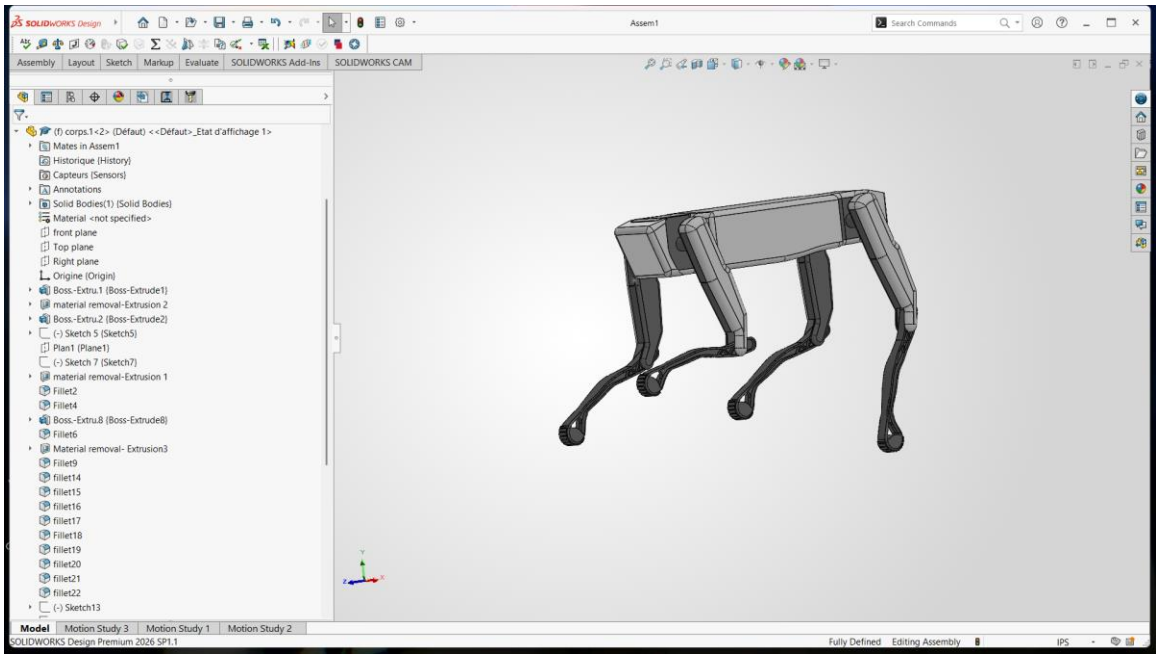
7. To validate the robot's movement through simulation and motion analysis.

8. To evaluate the stability and performance of the quadruped robot during locomotion.

9. To study the fundamentals of robotic locomotion, gait planning, and motion control.

10. To provide a foundation for future enhancements such as autonomous navigation, sensor integration, and advanced gait control algorithms..

Mechanical Structure Design



The mechanical structure of the quadruped robot was designed to provide stability, simplicity, and ease of motion analysis. The robot consists of a central rectangular body frame and four identical legs arranged symmetrically at the four corners of the body. This configuration ensures balanced weight distribution and improved stability during locomotion. Each leg is designed with two links: an upper leg link and a lower leg link. The links are connected using revolute joints that provide rotational motion. Every leg possesses two degrees of freedom (2-DOF), namely the hip joint and the knee joint. The hip joint enables forward and backward movement of the leg, while the knee joint allows lifting and extension of the lower leg during walking.

The body frame acts as the main supporting structure and serves as the mounting platform for all four legs. The design was created in SolidWorks using individual part models that were later assembled into a complete quadruped robot. Appropriate mates and constraints were applied to ensure proper alignment and realistic movement of the joints.

Main Components

- Rectangular Body Frame

- Front Left Leg Assembly

- Front Right Leg Assembly

- Rear Left Leg Assembly

- Rear Right Leg Assembly

- Hip Joints

- Knee Joints

Design Specifications

Parameter	Specification
Number of Legs	4
Degrees of Freedom per Leg	2
Total Degrees of Freedom	8
Body Shape	Rectangular
Joint Type	Revolute Joint
CAD Software Used	SolidWorks

Design Considerations

- Symmetrical leg arrangement for improved balance.

- Lightweight structure for efficient motion.

- Simple mechanism for easy manufacturing and assembly.

- Adequate joint motion range for gait generation.

- Stable body structure to support locomotion.

The final mechanical structure successfully provides the necessary mobility and stability required for basic quadruped locomotion and serves as a suitable platform for gait analysis and motion validation.

CAD Modeling and Assembly

The Computer-Aided Design (CAD) modeling of the quadruped robot was carried out using SolidWorks software. Individual components of the robot were designed separately and later assembled to form the complete quadruped structure. The CAD model was developed with a focus on simplicity, structural stability, and ease of motion analysis. The design process began with the creation of the main body frame, which serves as the central supporting structure of the robot. Subsequently, the upper leg and lower leg components were modeled according to the required dimensions and motion requirements. Each leg assembly was designed to provide two degrees of freedom through hip and knee joints. After creating all individual parts, the assembly process was performed using SolidWorks Assembly tools. Appropriate mates such as concentric mates, coincident mates, and limit angle mates were applied to ensure proper alignment and controlled movement of the joints. The leg assemblies were attached symmetrically to the body frame to maintain balance and uniform load distribution.

CAD Modeling Procedure

1. Creation of the rectangular body frame.
2. Modeling of upper leg links.
3. Modeling of lower leg links.
4. Creation of hip and knee joint connections.
5. Assembly of individual leg components.
6. Integration of all four legs with the body frame.

7. Application of mates and motion constraints.

8. Verification of assembly integrity and movement.

Assembly Features

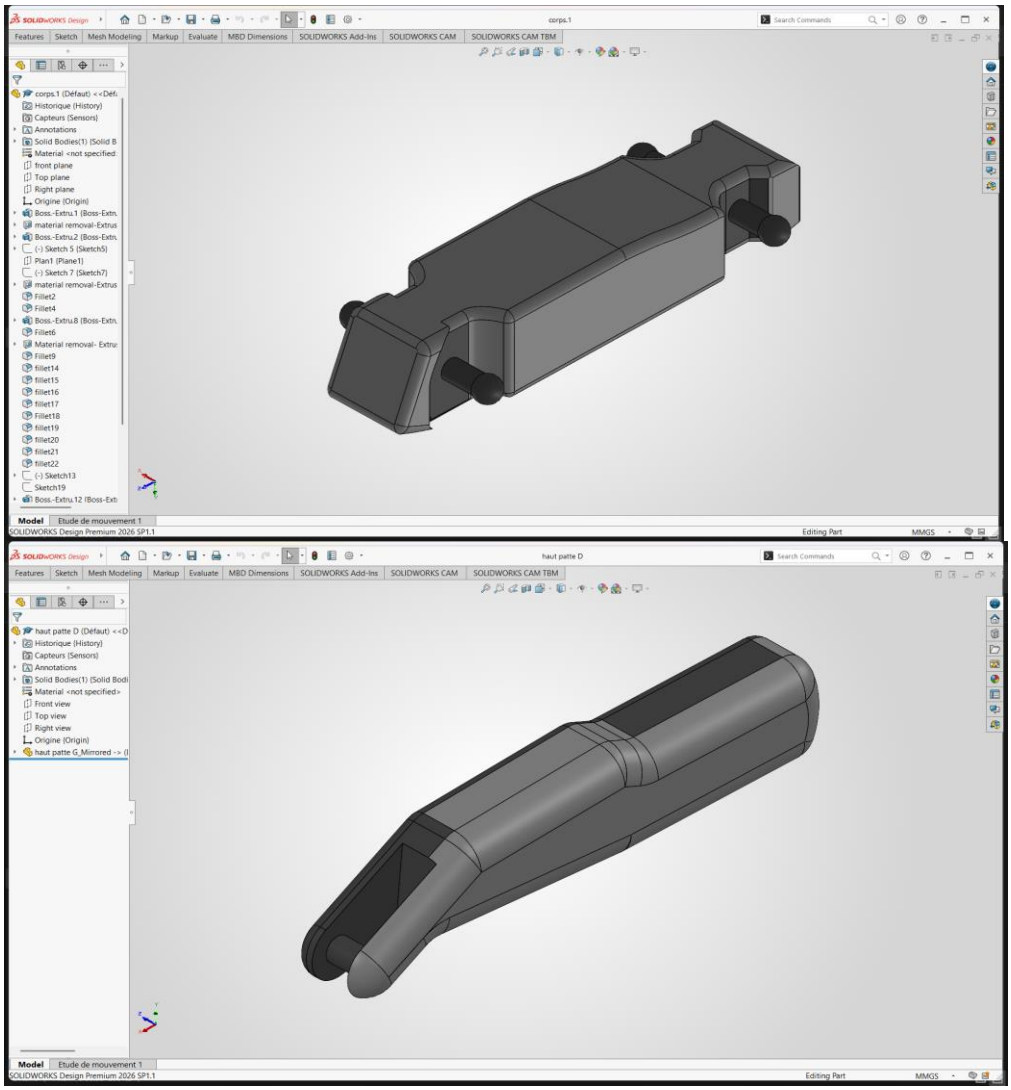
- Four identical leg assemblies.
- Symmetrical placement of legs.
- Revolute joint-based motion.
- Proper alignment using assembly mates.
- Controlled motion using limit angle constraints.

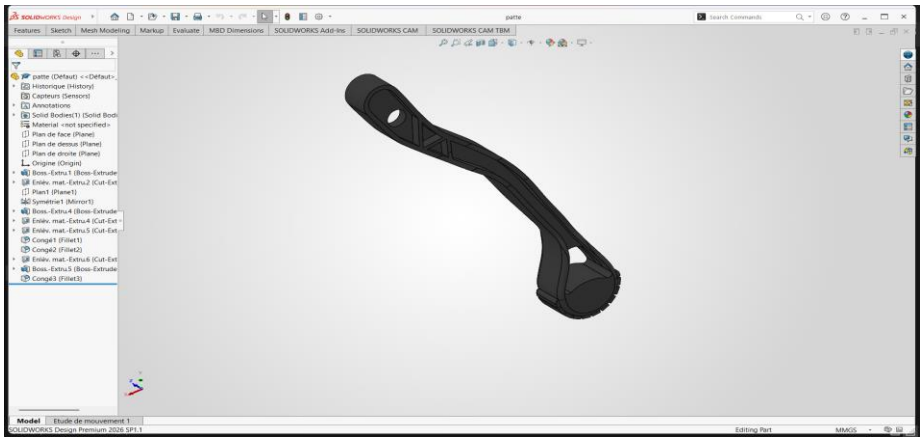
Software Used

Parameter	Details
CAD Software	SolidWorks
Modeling Type	3D Parametric Modeling
Assembly Type	Mechanical Assembly
Joint Type	Revolute Joint
Motion Constraints	Limit Angle Mates

Outcome

The completed CAD assembly successfully represents a functional quadruped robot capable of performing basic locomotion. The assembly allows controlled joint movement and serves as the foundation for gait analysis, motion validation, and stability studies.




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Mesh Modeling
Markup
Evaluate
MBD Dimensions
SOLIDWORKS Add-Ins
SOLIDWORKS CAM
SOLIDWORKS CAM TBM
Model
Etude de mouvement 1
Editing Part
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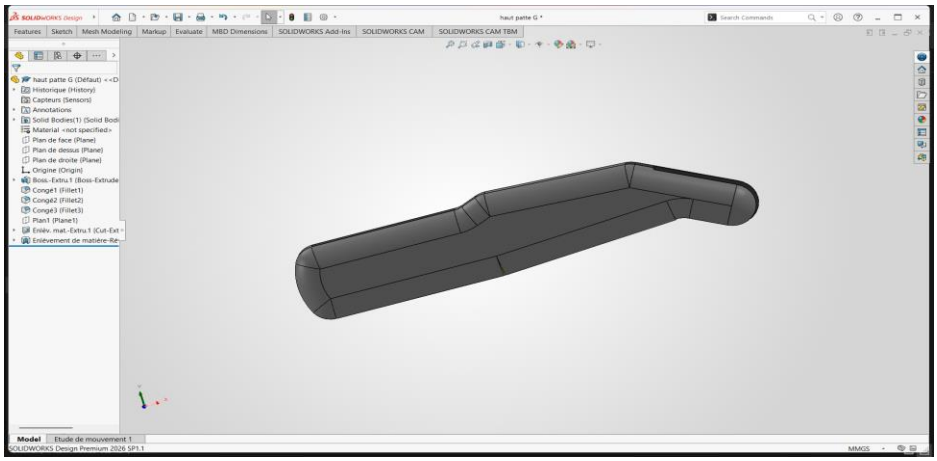

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Figure: Complete Assembly Model:

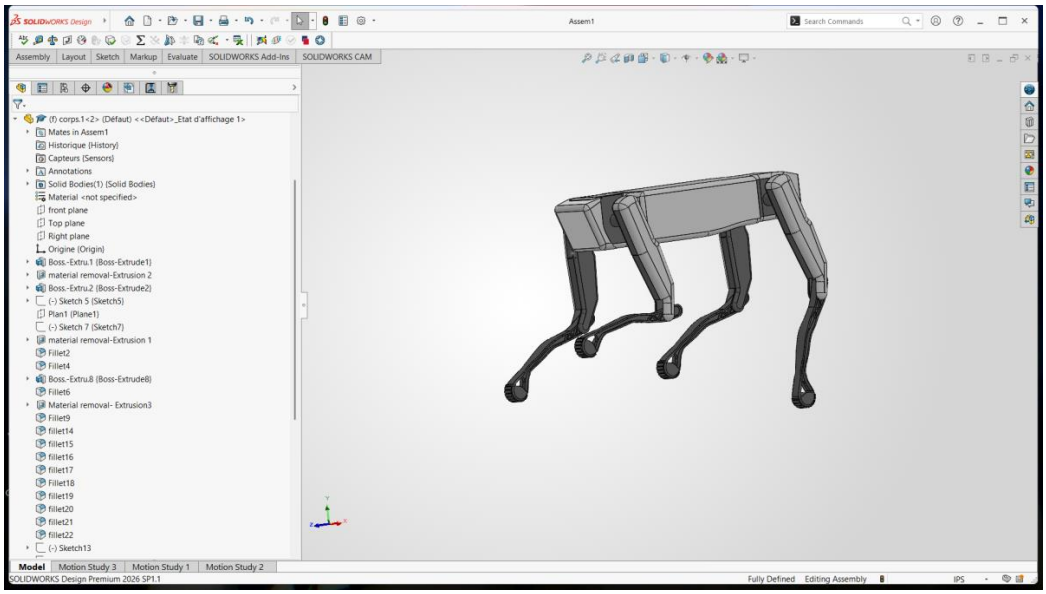
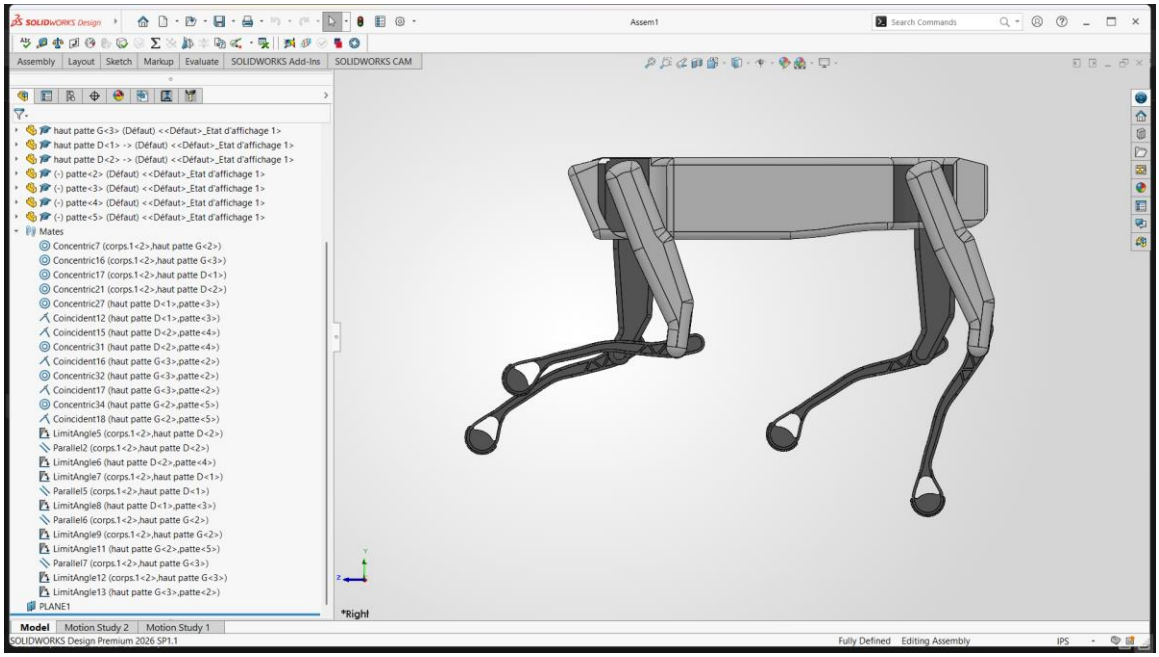


Figure: Assembly with Mates and Constraints:



Joint Definition and Motion Limits

The locomotion of the quadruped robot is achieved through rotational movement at the hip and knee joints. To ensure realistic and controlled motion, appropriate joint definitions and motion limits were assigned during the assembly process in SolidWorks.

Each leg of the robot consists of two revolute joints:

1. Hip Joint: The hip joint connects the upper leg link to the main body frame. This joint is responsible for moving the leg forward and backward during walking. The hip joint plays a major role in generating the stride length and controlling the gait cycle of the robot.

Motion Range

- Minimum Angle: -30°

- Maximum Angle: +30°

The selected range provides sufficient leg swing while preventing excessive rotation and mechanical interference between components.

2. Knee Joint: The knee joint connects the upper leg link to the lower leg link. It enables the leg to lift and extend during locomotion, allowing the robot to clear obstacles and maintain balance during movement.

Motion Range

- Minimum Angle: 0°

- Maximum Angle: 45°

The knee motion limit was chosen to provide adequate lifting action while maintaining structural stability.

Joint Constraints

To achieve controlled movement, Limit Angle Mates were applied in the SolidWorks assembly. These constraints prevent the joints from rotating beyond their specified limits and ensure smooth and realistic motion.

The following mate types were used:

- Concentric Mate

- Coincident Mate

- Limit Angle Mate

These mates define the allowable range of movement for each joint while maintaining proper alignment between components.

Joint Specifications

Joint	Function	Motion Range
Hip Joint	Forward and backward leg movement	-30° to +30°
Knee Joint	Leg lifting and extension	0° to 45°

Observations

- Smooth rotational motion was achieved within the specified limits.

- No major interference between components was observed.

- The applied constraints ensured realistic joint behavior.

- The defined motion limits are suitable for basic quadruped gait generation and locomotion studies.

The successful implementation of joint definitions and motion limits provides the foundation for gait planning, motion validation, and stability analysis of the quadruped robot.

Figure: Limit Angle Mates :

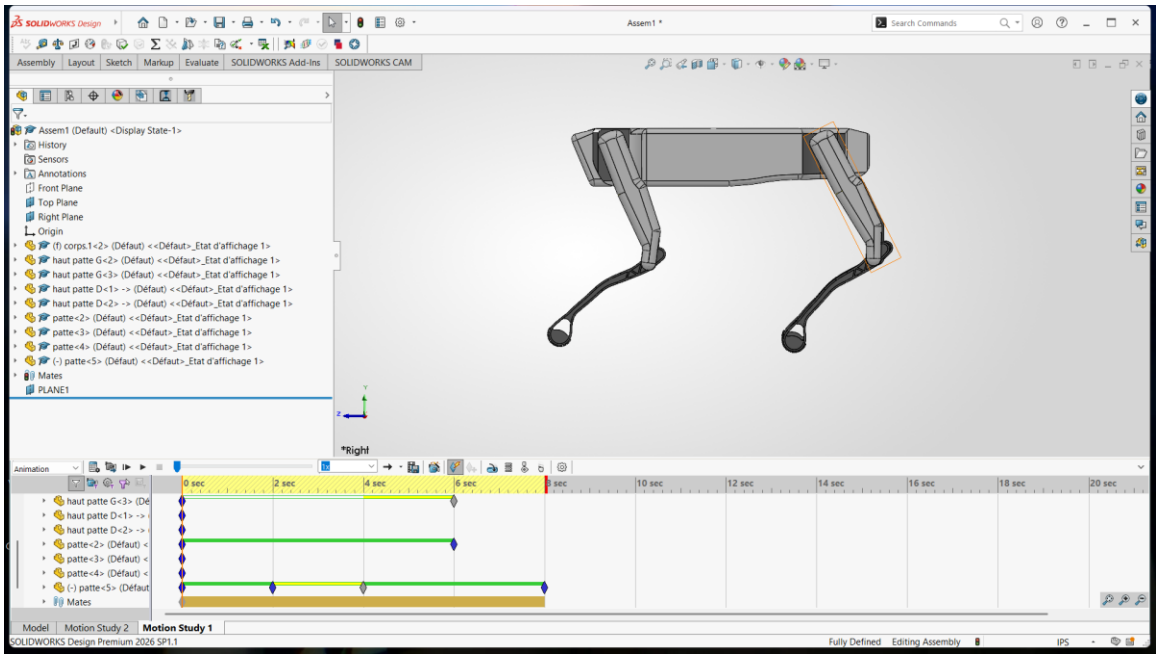
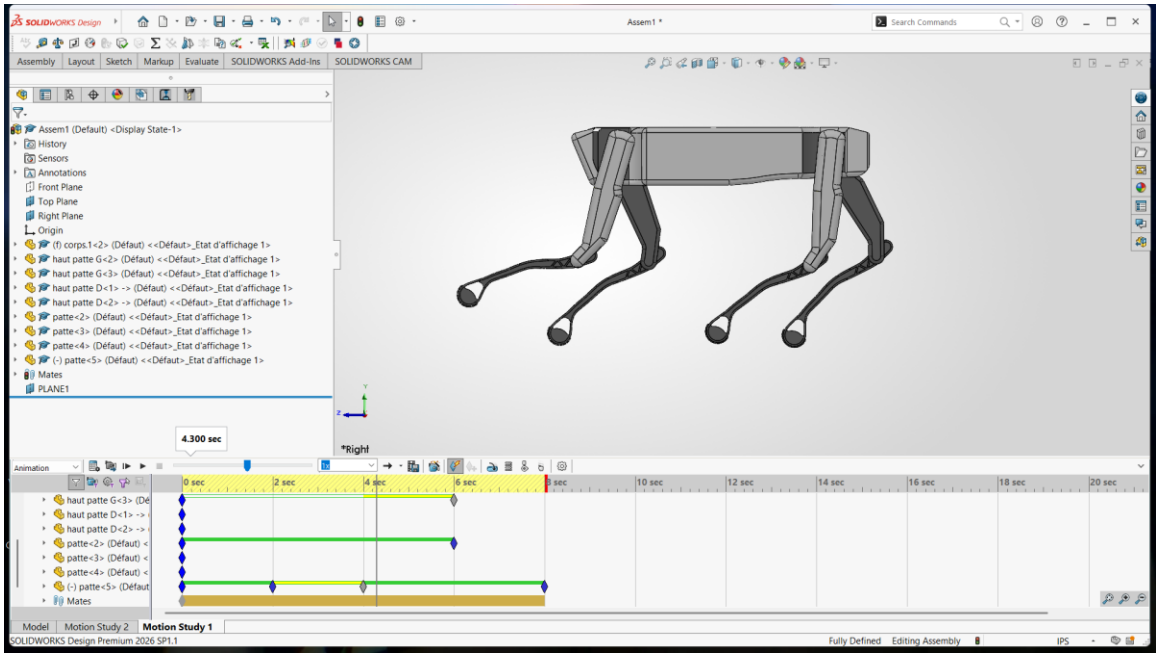


Figure: Hip Joint Motion Range:



Gait Cycle Calculation

A gait cycle represents the sequence of leg movements that enables the quadruped robot to walk in a stable and coordinated manner. In this project, a simple periodic gait pattern was considered to study the locomotion of the robot. The motion of the hip and knee joints was generated using sinusoidal functions, which provide smooth and continuous movement during walking.

The gait cycle duration was assumed to be 4 seconds, representing one complete walking cycle. During this cycle, the hip and knee joints move periodically to simulate the lifting, swinging, and placement of the legs.

Assumptions

- Gait Type: Simple Walking Gait

- Gait Cycle Time (T): 4 seconds

- Hip Joint Maximum Angle: ±30°

- Knee Joint Maximum Angle: 45°

- Smooth periodic motion is assumed.

- Effects of terrain irregularities are neglected.

Hip Joint Trajectory

The hip joint motion is represented by a sinusoidal function:

$\theta_h(t) = 30 \sin(2\pi t/T)$

where:

- $\theta_h(t)$ = Hip joint angle (degrees)

- T = Gait cycle period (4 seconds)

- t = Time (seconds)

The hip joint controls the forward and backward swing of the leg during locomotion.

Knee Joint Trajectory

The knee joint motion is represented by:

$\theta_k(t) = 45 \sin(2\pi t/T + \pi/2)$

where:

- $\theta_k(t)$ = Knee joint angle (degrees)
- $\pi/2$ = Phase shift between hip and knee motion

The phase shift ensures coordinated movement between the hip and knee joints, resulting in smoother walking behavior.

Gait Cycle Phases

The complete gait cycle can be divided into four stages:

1. Initial Contact Phase

- Leg touches the ground.
- Body weight begins to transfer.

2. Support Phase

- Leg supports the robot body.
- Stability is maintained.

3. Swing Phase

- Leg moves forward.
- Hip and knee joints rotate simultaneously.

4. Preparation Phase

- Leg prepares for the next ground contact.
- Cycle repeats continuously.

Sample Joint Angles

Time (s)	Hip Angle (°)	Knee Angle (°)
0	0	45
1	30	0
2	0	-45
3	-30	0
4	0	45

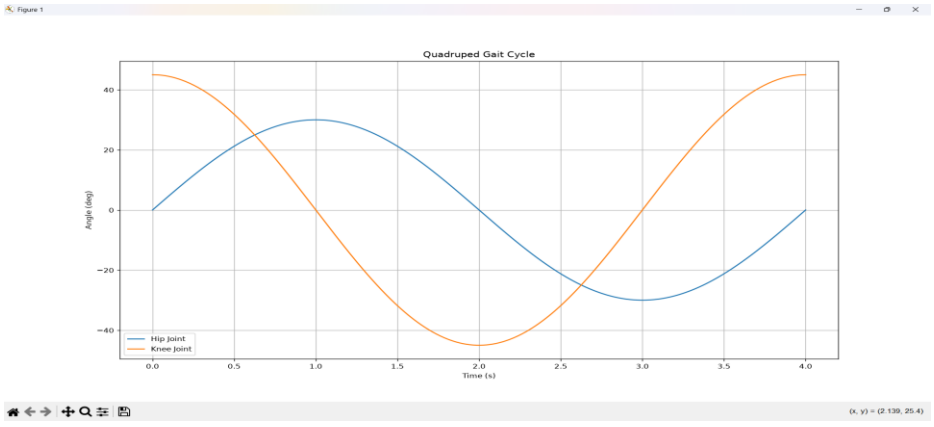
Results

The generated gait trajectory produces smooth periodic motion for both joints. The hip joint controls leg swing, while the knee joint controls lifting and extension. Together, these motions create a simplified walking pattern suitable for quadruped locomotion studies.

Figure: Python Code for Gait Calculation:



Figure: Hip and Knee Joint Angle Graph:



The calculated gait cycle serves as the basis for motion validation and stability analysis of the quadruped robot.

Future Improvements

Although the developed quadruped robot successfully demonstrates basic locomotion and gait generation, several improvements can be made to enhance its performance, stability, and functionality.

1. Increase Degrees of Freedom (DOF)

The current design uses only two degrees of freedom per leg. Additional joints can be incorporated to improve flexibility, maneuverability, and obstacle-climbing capability.

2. Implementation of Advanced Gait Algorithms

More sophisticated gait patterns such as trot, pace, bound, and gallop can be implemented to improve walking speed, stability, and adaptability to different terrains.

3. Integration of Sensors

Sensors such as IMU (Inertial Measurement Unit), ultrasonic sensors, force sensors, and encoders can be integrated to provide real-time feedback and improve balance control.

4. ROS-Based Control System

The Robot Operating System (ROS) can be incorporated for advanced motion planning, sensor integration, communication, and autonomous navigation.

5. Autonomous Navigation

Future versions of the robot can be equipped with path-planning and obstacle-avoidance algorithms, enabling fully autonomous movement in unknown environments.

6. Artificial Intelligence and Machine Learning

AI-based techniques can be used to optimize gait patterns, improve terrain adaptation, and enable intelligent decision-making during locomotion.

7. Real-Time Control Hardware

Microcontrollers and embedded systems such as Arduino, ESP32, Raspberry Pi, or STM32 can be integrated to control servomotors and execute real-time walking motions.

8. Improved Mechanical Design

The mechanical structure can be optimized by using lightweight materials, stronger joints, and compact actuators to increase efficiency and reduce power consumption.

9. Dynamic Stability Analysis

Future work can include dynamic stability studies considering body motion, external disturbances, and uneven terrain conditions for more realistic performance evaluation.

10. Physical Prototype Development

A complete hardware prototype can be fabricated and tested to validate the simulation results and evaluate real-world performance.

Summary:

The proposed quadruped robot provides a strong foundation for future research and development in robotic locomotion. By integrating advanced control techniques, intelligent algorithms, and real-time hardware systems, the robot can be transformed into a fully autonomous and highly capable legged robotic platform..

Conclusion

In this project, a simplified quadruped robot was successfully designed, modeled, and analyzed using CAD and simulation tools. A complete 3D model of the robot was developed in SolidWorks, consisting of a rectangular body frame and four identical legs with two degrees of freedom per leg. Appropriate joint definitions and motion limits were applied to achieve realistic leg movement and ensure smooth operation.

A basic gait cycle was generated using mathematical trajectory equations, and Python was utilized to plot the hip and knee joint angle variations throughout the walking cycle. The generated trajectories demonstrated smooth and coordinated motion suitable for quadruped locomotion. Motion validation and stability analysis confirmed that the proposed design is capable of maintaining balance while performing basic walking movements.

The project provided valuable knowledge in robotic locomotion, CAD modeling, joint motion analysis, gait generation, and simulation techniques. It also highlighted the importance of proper mechanical design and motion planning in the development of legged robots.

Overall, the objectives of the project were successfully achieved, and the developed quadruped robot serves as a strong foundation for future work involving advanced gait control, autonomous navigation, sensor integration, and intelligent robotic systems..